DATA FLOW DIAGRAMS (DFD)

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. DFD Level 0: Context Diagram

The Level 0 context diagram represents the boundary of the OptiCrop system. The primary actor (User) interacts directly with the system:

- Input Data: The User inputs N, P, K nutrients, Soil pH, local temperature, humidity, rainfall, region, and season details.
- Output Data: The system processes input configurations and displays the suggested crop, classification confidence, yield forecast, soil cluster profile, and NPK corrective prescriptions.

2. DFD Level 1: System Process Diagram

The Level 1 DFD decomposes the system into core components that handle data processing and persistence:

- Process 1.0 (Input Sanitizer): Validates and checks user-entered numbers for physical validity (e.g. pH between 0 and 14).
- Process 2.0 (Data Scaling): Standardizes features using the saved scaler model.
- Process 3.0 (ML Model Inference): Feeds scaled features to the classifier, regressor, and K-Means models.
- Process 4.0 (Prescription Generator): Compares inputs to standard crop requirements and outputs corrective measures.
- Process 5.0 (Data Storage): Appends inputs and output records to the local CSV logger for future model training.